Maven Music Customer Churn Analysis (Scoping the project)

The Situation

I have been hired as a Junior Data Analyst for Maven music, a streaming service that's been losing more customers than usual the past few months and would like to use data science to figure out.

How to reduce customer churn?

(Major issue: Not less customers but Monthly Revenue growth is slowing)

The Assignment

I will have access to data on Maven music's customers including subscription details and music listening history.
My task is to Gather, clean and explore data to provide insights about the recent customer churn issues And then prepare it for modeling in the future.

The objective

1. Scope the data science project
2. Gather the data in Python
3. Clean the data
4. Explore and visualize the data
5. Prepare the Data for modeling

In this particular project I'll be focusing on **how to scope a project**. I'm going to be preparing data to be input into a model but not going to do the modeling now.
In the real world, starting with data prep and EDA are the most important parts of modeling and I want to showcase that here.
If a project isn't properly scoped, then a lot of time can be wasted solving the wrong problem or going down the wrong path.

Scoping the Project:

Step 1: Think like an end user

When introduced to a new project- start by asking questions that help you:

- **Emphasize** with the end user - what do they care about?
- Focus on **impact** - what metrics are important to them?

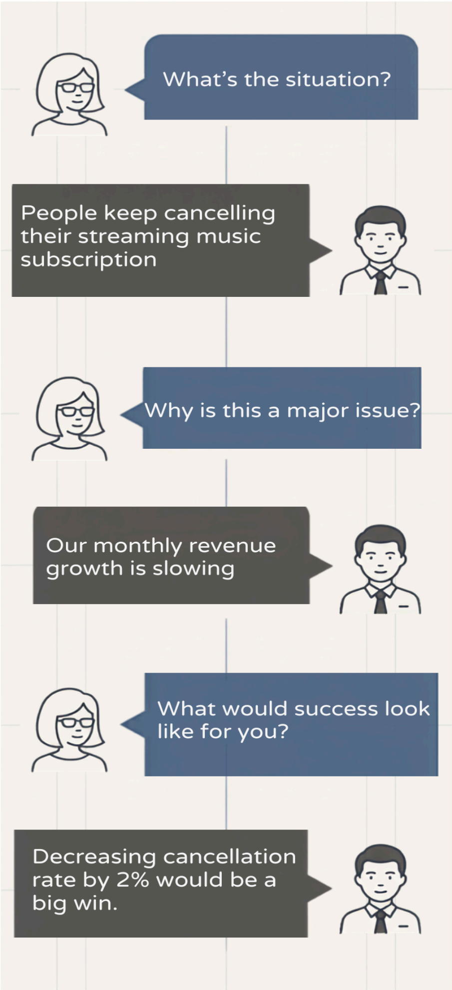
Step 2 : Find the problem

Here I am considering why our customers are opting out of the subscription.

There can be many reasons Some of them are:

1. Maybe our product isn't really differentiated from our competitor.
2. Maybe there are a lot of technical bugs and that's causing them to leave.
3. Maybe our product is too expensive.
4. Maybe there's not enough music in the library.
5. Do we even know why our customers are leaving?

The Problem: Do we even know why our customers are leaving?



Step 3 : Brainstorming Solutions

Once we have our problem, the next step is to brainstorm our solutions.

Some Potential solutions:

- Ask the product team to add a survey to capture the cancellation feedback.
- Speak with customer representative about any changes they have noticedIn the recent customer interactions.
- Conduct customer interviews to gather qualitative data and insights.
- Research external factors that might impact cancellations Example competitive landscape, news Etc
- Use dataTo identify the top predictors for account cancellation.

The Solution: Use data To identify the top predictors for account cancellation.

Step 4 : Determine the techniques

The main question we're trying to answer is, is this a supervised learning problem or an unsupervised learning problem?

- for supervised learning.

we use historical data to predict the future and

- for unsupervised learning,

We find patterns or relationships in the data.

Main Question: Do we need supervised or unsupervised learning?

Since our solution is to use data to identify the top predictors for account cancellation we will be needing historical data and hence going for supervised learning.

The Technique: Supervised learning.

Supervised learning:

Now that we have our technique finalized we need to figure out **how to structure our data**. Since we are using a Supervised learning model here we need to have data that has a label**.**
A label is an observed variable which we are trying to predict.

Customer	Months Active (X)	Monthly Payment (X)	Listened to Indie Artists? (X)	Cancelled? (Y)
Aria	25	$9.99	Yes	No
Chord	2	$0	No	No
Melody	14	$14.99	No	Yes

- **Target Variable (Y):** `Cancelled?`
- **Features (X):** Subscription history, monthly payments, listening behavior, demographics.

We take this data set here and put it into a predictive model and once we have created your model, when we have a new customer with all their information like

- how many months they have been active
- how much they are paying each month

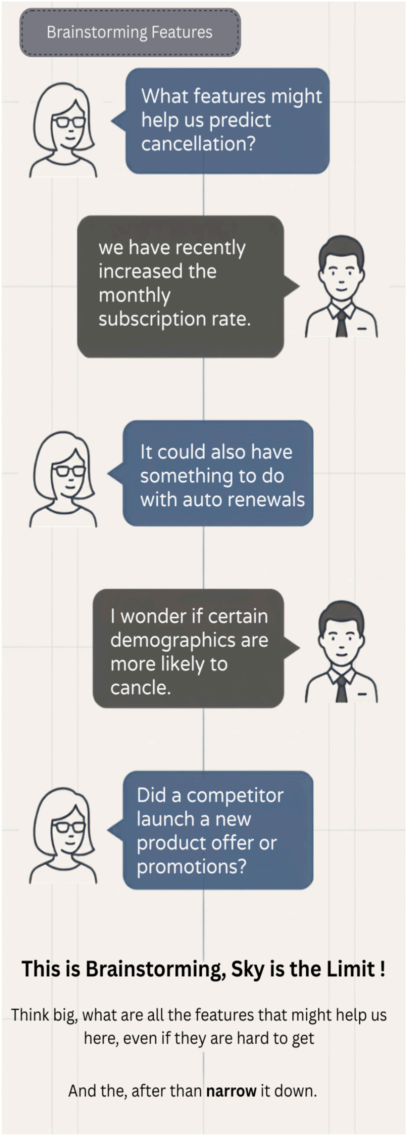
Then we can predict whether they are going to cancel the subscription or not.

Step 5: Model Features

Here we identify features that are good for our analysis. A feature is known as an input into a model. In our case Supervised learning feature that will do a good job predicting cancellation

- Can be Looking at past information on what customer listened Or
- maybe looking at how muchThey spend each month

It's important when we are scoping to start brainstorming these features because then we'll know which data we need to gather for analysis.



After we have brainstormed features and thought about all the different inputs we can put into our model, the next step is to figure out the data sources or where we are going to get our data.
Features with sources:

Features:	Potential Sources:	Ease of Access:
Monthly rate	Customer subscription history, customer listening history	Easy *(internal data)*
Auto-renew		
Age	Customer demographics	
Urban vs. Rural		
Competitor Promotion	Customer's other subcriptions, competitor promotion history	Hard *(external data)*

Step 6: Data sources

Once we have all of this on the page The next step is to think about how easy or hard it is to get access to all of these data sources.
It's hard to get everything we have in mind so we need to start with the easier data sources with which we can produce great features.
And then work on the analysis, we can continue to add on additional data sources later on.

Step 7: Data scope

Till now we have decided on our data, talked about data structures, finalized the features and sources. Now it's time to scope it all down.

Remember: More data doesn't necessarily mean a better model.
Quality over quantity !

In this case we will be focusing on customer subscription history and customer listening history. These are internal data sources easy to get our hands on quickly.

Then we need to focus on filters i.e Time frames. **How many months back(No correct answer).**

So let's say we have a lot of data in the warehouse and we are going to look into the data from the **past three months** from customers with only customers who are actively subscribed.

These are the decisions made based on our conversation with the end user/ Stakeholder .

Pro tip: Aim to produce a **minimum valuable product (MVP)** At this stage. Going deep Without feedback can Head down to Productive path or over complicating the solution.

So start with a small subset like three months of customer subscription And listening history.
If they are not good as expected then we could look into more data sources or longer time frames.

Have talked about thinking like an end user, Brainstorming problems and solutions figuring out the modeling techniques that we plan to use as well as the data requirements. Final step is to summarize everything we have talked about, project scope and objectives:

- What techniques and data do we plan to leverage?
- What specific impact are we trying to make?

Step 8: Summarize the scope and objective(Guiding star)

“We plan to use **Supervised Learnings** to predict which customers are likely to cancel their subscription using **the past three months of subscription and listening history.**
This allows us to:

- Identify the **top predictor for cancellation** and figure out how to address them
- Use the module to **flag customers who are likely to cancel** and take proactive steps to keep them subscribed.

Our Goal is to **reduce cancellation by 2% over the next year.”**